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The Palaeomagnetism of Precambrian Igneous and Sedimentary Rocks of the Orange River Belt in South Africa and South West Africa

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Summary

Palaeomagnetic results are presented for a variety of Precambrian rocks from south-western Africa. Oldest rocks studied are the Orange River Lavas (> 1850 My) from which the three least altered sites give a VGP at 254° E, 19° N near the polar wander curve for Africa at *c.*2100 My. Five sites from basic intrusions believed to be about 1200 My in age at O'okiep, Namaqualand, give a pole at 155° E, 15° S. Nineteen sites in lavas of the Barby Formation (Sinclair Group, < 1200 My) in South West Africa yield a significant direction after cleaning with five sites reversed with respect to the remainder; grouping is improved after correction for folding (> 1020 My). Palaeomagnetic results are also given for lavas of the overlying Guperas Formation (< 1020 My), sediments of the succeeding Auborus Formation, basic dykes at Klein Karas (878 ± 41 My), and six dolerite dykes from Namaqualand (probable age of 653 ± 70 My). Nine sites from the Fish River Formation (probable Lower Cambrian age) give pole positions correlating with a late Precambrian to Ordovician polar shift recognized from other Gondwanaland data. The Doornpoort Formation of Central South West Africa gives a direction of magnetization which a fold test dates as originating from the Damaran folding (550–500 My).

The results define a loop in the polar wander curve for Africa between 1200 and 1000 My which can be matched in size, shape and age to a loop in the North American curve implying that the two regions were in juxtaposition at this time.

1. Introduction

Palaeomagnetic studies of African Precambrian rocks were concentrated during the last decade on rocks of the Rhodesia/Kaapvaal (Transvaal) cratons, and have been summarized by McElhinny *et al.* (1968) who define an apparent polar wander (apw) curve for the intervals 2700–2300 My and 1970–1300 My. From late Precambrian palaeomagnetic results from Tanzania (Piper 1972) and Kenya (Brock, Raja & Vise 1972), Brock & Piper (1972) proposed an apw path to connect the Precambrian and Phanerozoic paths. Uncertainty surrounded this interpretation because there still remained large gaps in time coverage of the data and it was uncertain whether later tectonic activity had caused large relative movements between the cratonic regions of

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central and southern Africa where the separate portions of the apw curve were derived. This study is concerned with pre- and post-orogenic igneous rocks from the northern part of the Namaqualand Complex incorporating a region of varying structural trends but yielding a preponderance of age dates around 1100 My (Nicolaysen 1962; Clifford 1968) known as the Orange River or Namaqualand Belt; it abuts against the western margin of the Rhodesia–Kaapvaal craton. The study gives several new palaeomagnetic poles which help to resolve these uncertainties.

2. Geological background

Precambrian rocks of the region outcrop below Karroo and younger rocks in a strip of country running between the coast and 300–400 km inland (Fig. 1). Oldest rocks form a basement of metasediments and metavolcanics collectively known as the Kheis Group and intruded by large masses of granite-gneiss (Martin 1965; Haughton 1969; Joubert 1971). The Orange River Lavas sampled for this study at Vioolsdrif belong to this Group and are intruded by granite-gneiss giving an age of 1850 ± 50 My (de Villiers & Burger 1967) although they may be considerably older because granite intruding the Marydale Series of the Kheis in the type are given an age of 2600 My (Nicolaysen & Burger 1965).

In the western part of the southern South West Africa folded and largely unmetamorphosed rocks lie on probable Kheis basement subdivided as the Kunjas, Sinclair and Auborus Series by Beetz (1923) who grouped them as the Konkip System. A predominantly volcanic formation below the first series was subsequently recognized as the Nagatis Formation (Martin 1965) and was intruded by granites including the Tumuab granite (1290 ± 30 My) before deposition of the predominantly-clastic Kunjas series. Radiometric data suggest a possible age of 1200 My for acid volcanics of the Nagatis Formation (von Brunn 1969). The overlying Sinclair Group is predominantly volcanic and granites intrusive into it but predating the overlying Auborus Formation, include the Rooikam granite dated at 1020 ± 80 My (von Brunn 1969). At the top of the Sinclair Group and separated from it by angular unconformity is a suite of rocks including lava flows and sediments distinguished (Martin 1965) as the Guperas member (now considered to be a formation). These lavas were erupted after the bulk of the folding of the Sinclair Group was completed (B. R. Watters 1971, private communication).

Folding and intrusion of the Sinclair series was followed by deposition of a thick succession of conglomerates, breccias, red sandstones, and siltstones known as the Auborus formation. In the type area this formation was laid down in a N–S synclinal basin with thick conglomerates at the base and characterized by large lateral changes in lithology and thickness. Sedimentological studies (Miller 1969) suggest deposition under strongly oxidizing conditions with frequent exposure and formation of dessication features. Mild deformation and block faulting of the Auborus formation took place soon after deposition and was followed by erosion to the flat surface on which the overlying Nama Group was deposited. This Group comprises shallow water shelf sediments divided into three formations with which this paper is concerned only with the top (Fish River) Formation. The Nama is predominantly arenaceous or argillaceous but includes limestones and glacial horizons (Kröner & Rankama 1972). The Fish River Formation is a very shallow water sandstone and siltstone succession deposited in strongly oxidizing conditions. Germs (1972) suggested a Cambrian age for these beds on account of the presence of the trace fossil *Phycodes pedum.*

In the north-central part of South West Africa rocks belonging to the Damara System have a NE–SW Structural trend acquired during the Damaran episode of the Pan African orogeny which cuts across the trends imposed by earlier episodes to the south and north (Fig. 1). The Damara System is underlain, locally unconformably,

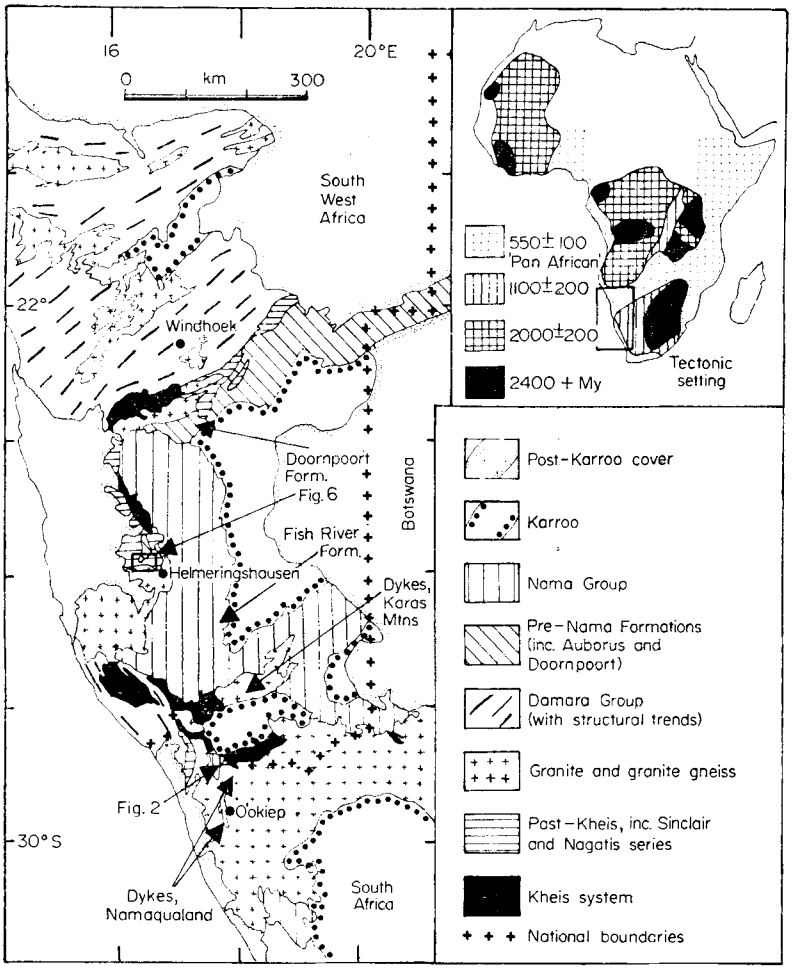


FIG. 1. Simplified geological map of south western Africa based mainly on Haughton (1969) and showing the location of the rock units sampled for this study. The inset diagram shows the region in relation to the major structural divisions of the African Precambrian based on Clifford (1968). The blank area in northern Africa outside of the Atlas region is left blank due to lack of data, but is currently thought to be crust in excess of 3000 My in age on which the 'Pan African' thermal event has been overprinted (Dr P. W. G. Tanner 1973, private communication).

by the arenaceous and argillaceous Nosib Group (Kröner 1971) which, along the southern margin of the Damara belt, unconformably overlies sediments previously described as the Tsumis Formation (Martin 1965) and now grouped into five separate units (Schalk 1970); of these the Doornpoort Formation has been sampled in the present study.

3. Palaeomagnetic results

3.1 *Field and laboratory techniques*

Except for the dykes of Klein Karas which were lump sampled, all the palaeomagnetic samples were collected using a portable coring drill (Doell & Cox 1967).

Up to seven cores were sampled at each site and oriented by sun sights. They were cut into inch-long cylinders in the laboratory, and one or two pilot samples were selected from each site for stepwise af demagnetization. As a rule steps of 100 Oe up to 1000 or 1600 Oe were chosen and samples were measured before and after treatment with a PAR Model SM-2D spinner magnetometer. A stability index (Briden 1972) was computed for each step and an optimum alternating field chosen from the behaviour of the pilot for treatment of the remaining samples from the site.

3.2 *Orange River Lavas, Vioolsdrif, Cape Province*

Palaeomagnetic sites were taken through a pile of lavas dipping to the north of NNW at angles which increase from 25° in the south to 50° in the north of the outcrop. The lavas are thick and between-lava sediments are largely absent. In stratigraphic sequence through the succession, sites are 46, 47, 112, 111, 110, 109, 51, 50, 48 and 49, the lowest site (46) being 20 m above the intrusive contact with granite gneiss (Fig. 2). The Orange River lavas are included in the Wilgenhout Drift Formation of the Kheis (de Villiers & Söhnge 1959) but a correlation with the Ventersdorp System should also be considered (J. H. W. Ward and A. Kröner 1972, private communications 1972).

Samples (total NRM intensities of magnetization in the range 3×10^{-7} and 3×10^{-4} G with geometric mean of $1 \cdot 2 \times 10^{-5}$ G) show moderate or low stability to af

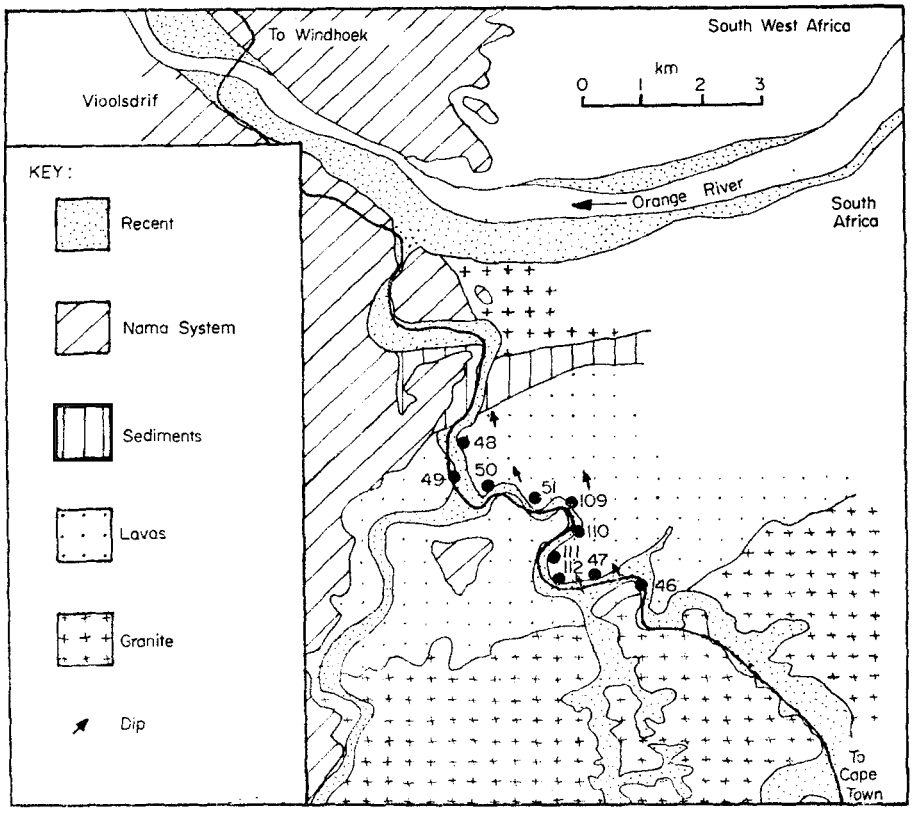


FIG. 2. Map of geology and sampling sites in the region south-east of Vioolsdrif, South Africa. The location of this region is shown in Fig. 1.

demagnetization. Those from sites 47 and 51 exhibit systematic and nearly exponential fall in their moments on demagnetization (Fig. 3(b)) with little change in direction of the remanence vector. Samples from sites 109, 111 and 112 have a total Natural Remanent Magnetism (NRM) with westerly declinations and low inclinations, or move towards this direction after demagnetization; the pilot from site 112 shows no decrease in moment or change in direction on demagnetization to 1000 Oe (Fig. 3(a) and (b)). Sites 50 and 110 show very unstable behaviour on demagnetization. (Fig. 3(b)) characterized by an irregular fall of their moments. They are not significantly grouped at the 95 per cent confidence level either before or after demagnetization although three of the five samples from site 50 move close to the direction exhibited by site 51 after demagnetizing in fields between 200 and 500 Oe. Three samples from site 47 give a close grouping which is maintained after demagnetization and show exponential fall in moments; the other three samples of this site give scattered directions both before and after cleaning although vectors move towards that of the three stable samples which are used in computation of the site mean. Site 49 has closely grouped total NRMs with a mean near to that of site 46. On demagnetization however, the vectors become scattered but three of the samples move to westerly negative inclinations near to that shown by site 51. After cleaning sites, 46, 48 and 51 are moderate to well grouped, 47, 109, 111 and 112 are less well grouped and site 49 is significant before cleaning; three samples from each of sites 49 and 50 are well grouped after cleaning (Table 1).

There is systematic change in direction of magnetization through this succession (Figs 2 and 4). Sites near the base and top have easterly negative directions while sites near the middle have westerly declinations with shallow positive or steeper

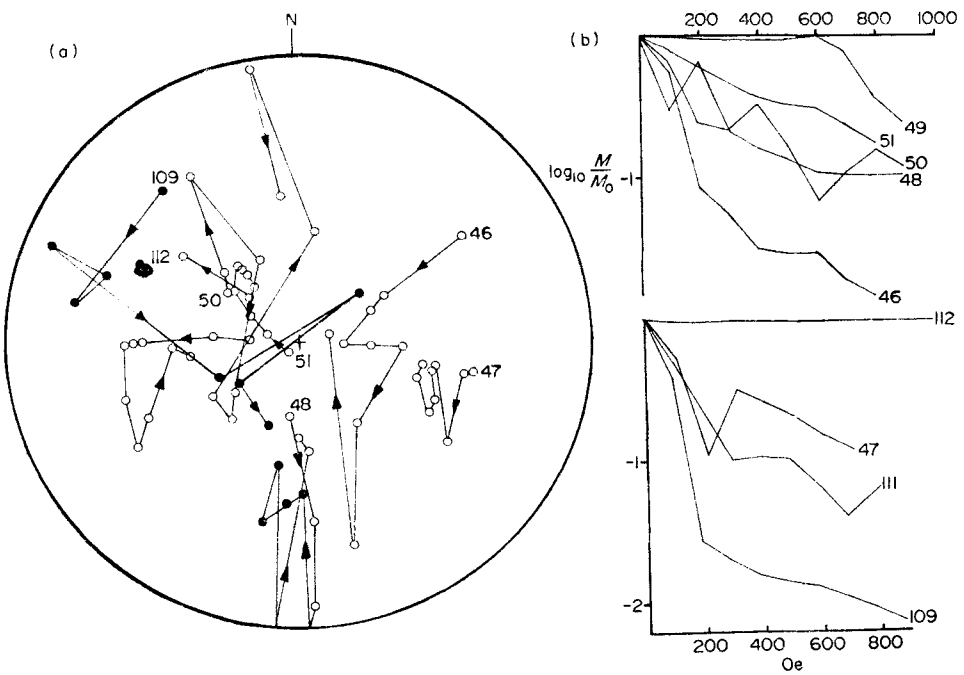


FIG. 3. Behaviour of (a) directions of magnetization, and (b) magnetic moments of some pilot samples from the Orange River lavas near Vioolsdrif, South Africa, with progressive af demagnetization in steps of 100 Oe. Throughout this paper open symbols refer to upper hemisphere plots and closed symbols refer to lower hemisphere plots.

Table 1

Orange River Lavas, Vioolsdrif, Cape Province, corrected for dip

Site No.	NRM site mean statistics						Cleaning field (Oe)	Cleaned directions, site mean statistics						Pole position	
	N	D	I	R	k	α_{95}		N	D	I	R	k	α_{95}	°E	°N
46	5	69	−49	4·8	22·3	17	200	5	54	−49	4·8	22·3	17	123	44
47	5	99	−26	3·6	7·4	36	200	4	101	−23	3·6	8·3	33	123	−4
112	6	321	7	5·5	9·4	23	300	6	321	7	5·4	8·5	24	321	41
111	4	132	11	3·7	8·8	33	300	4	218	19	3·7	10·7	30	267	−50
110	6	—	—	2·4	—	—	—	—	—	—	—	—	—	—	—
109	5	268	−41	3·7	3·0	53	300	5	286	13	4·2	5·3	36	302	11
51	5	261	−71	3·0	2·0	74	300	5	278	−54	4·8	24·6	16	259	22
50	5	—	—	2·6	—	—	300	3	252	−56	2·8	8·0	46	247	4
49	5	75	−45	4·0	4·1	43	300	3	289	−59	2·8	12·0	38	257	32
48	5	148	−64	4·8	18·1	18	100	5	166	−31	4·6	9·1	27	180	−42
(i) Sites 50, 49 (cleaned) and 50, after tilt correction								3	273	−57	3·0	59·1	16	254	19
(ii) Sites 46, 49 (uncleaned) and 47 after tilt correction								3	61	−61	2·9	13·9	34	140	40

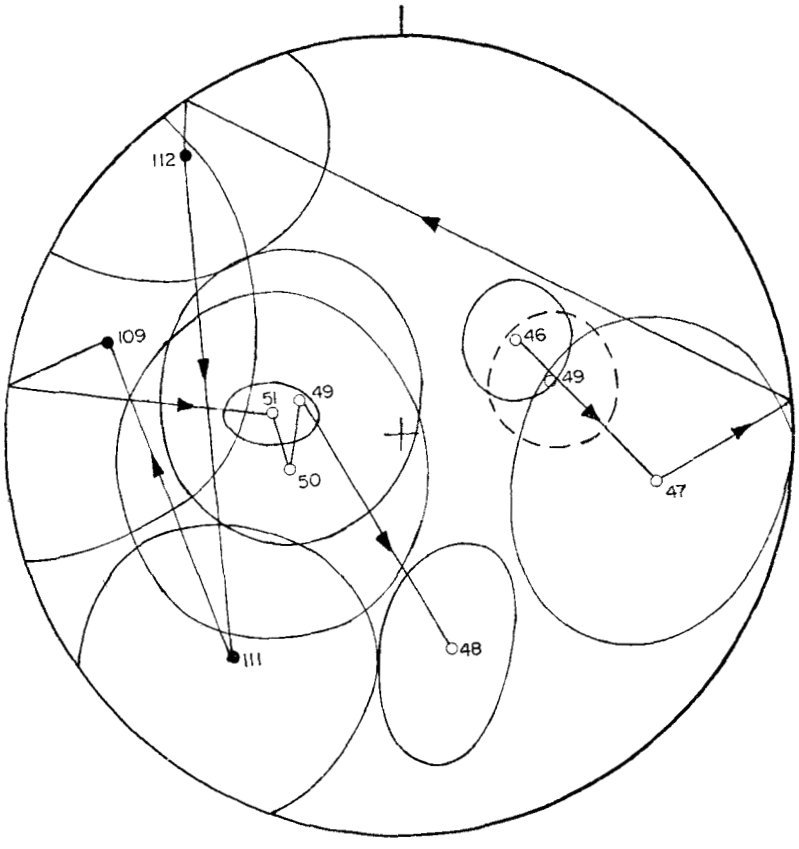


FIG. 4. Group mean directions of magnetization for the sites in the Orange River lavas at Vioolsdrif with 95 per cent confidence circles. The direction of site 49 is shown before (dashed circle) and after (full circle) cleaning; all the other directions are cleaned.

negative inclinations. Since a precise reversal of magnetization is not observed, the two clear directions of magnetization to emerge from the study probably merit interpretation in the context of variable amount of remagnetization of the lava sequence. The direction obtained from site 51 survives demagnetization to high fields and is supported by the directions obtained from the majority of samples from sites 49 and 50 after cleaning. These three directions give a combined mean (Table 1) of $D = 311°$, $I = -27°$ before tilt correction and $D = 273°$, $I = -57°$ after tilt correction. Site 46 gives a mean direction of $D = 69°$, $I = -49°$ which survives after demagnetization to 600 Oe although the grouping is not so good as site 51 and tends to deteriorate with cleaning. The total NRMs of site 49 are near this direction and three samples from site 47 give a similar direction both before and after cleaning in fields between 300 and 600 Oe. These three sites give a mean direction of $D = 61°$, $I = -61°$ ($D = 96°$, $I = -45°$ before tilt correction), and since this direction is best defined by total NRMs, it is regarded as a low coercivity component almost certainly secondary with respect to the directions derived from sites 50, 51 and 49 (cleaned) which is best defined after demagnetizing in high fields. Interpretation of these directions is considered in Section 4.1.

These lavas are locally metamorphosed to upper greenschist/lower amphibolite facies and least altered are sites 109–111 with pyroxene marginally altered to actinolite

and the feldspar patchily sericitized. Content of opaque minerals is small with pyrite predominant and no resolvable oxidation in few irregular grains of titanomagnetite present; they are marginally granulated. Sites 49 and 50 retain some pyroxene and have primary igneous texture better preserved, although groundmass is largely chlorite, amphibole and sericitized feldspar. Small amounts of unoxidized titanomagnetite present nucleate chlorite, and have alteration rims including rutile and haematite. Sites 46 and 47 are the most strongly altered sites of the collection with primary textures obliterated by secondary amphibole, quartz and feldspar; aggregates in the groundmass, apparently mainly of rutile may have pseudomorphed titanomagnetites presumably with the iron incorporated in amphiboles. The petrographic examination does show that sites giving the stable direction near $D = 273°$, $I = -57°$ are less altered than the sites giving a stable direction near $D = 96°$, $I = -44°$. Site 46 is also veined by at least two phases of quartz-carbonate mineralization, and the magnetization of sites 46 and 47 may be later than that of sites 50 and 51, dating from the time of metamorphism although the carrier of the secondary magnetization is not obvious from petrographic examination.

3.3 *Basic intrusions, O'okiep, Cape Province*

In the O'okiep region of the Namaqualand Copper Field metamorphosed sedimentary rocks possibly belonging to the Kaaien Formation of the Kheis (Haughton 1969) have been folded along east–west axes and intruded by later granites and copper-bearing norites. These latter bodies are associated with sharp folds superimposed on the regional foliation and are confined to certain levels in the stratigraphy (Benedict *et al.* 1964). The O'okiep mining company have kindly provided the author with samples drilled from single surface sites in five of these basic bodies (A. F. Lombaard & J. A. Muller 1971, private communication). The bodies are: Wheal Heath (site 34), Kliphoog (35), Fonteintjie (36), Hoits (37) and Ring Dyke (38). Two age determinations of 1010 My and 1060 My with a preferred mean of 1042 My have been made on these rocks (Nicolaysen & Burger 1965). According to Dr A. Kröner (private communication 1972) this age probably reflects a strong thermal episode associated with extensive and large scale lateral shearing in this region; new work suggests a slightly older age (Professor F. N. Clifford 1972, private communication).

Table 2

O'okiep basic intrusions, Namaqualand, South Africa.
Cleaned directions, site mean statistics

Site No.	Cleaning field (Oe)	No. of sites	D	I	R	k	α_{95}	Palaeomagnetic pole position °E	°S
34	400	5	146	−66	4·59	9·7	26	176	5
35	400	4	132	−45	3·98	148·7	8	154	17
36	400	5	133	−26	3·86	3·5	48	144	27
37	1000	4	125	−51	3·85	19·4	21	152	9
38	500	4	126	−52	3·97	97·1	9	154	9
			Group mean statistics, cleaned directions						
		5	131·6	−48·4	4·86	27·8	14·8	155·4	15·0

($d\Psi = 12·7$, $d\chi = 19·4$)

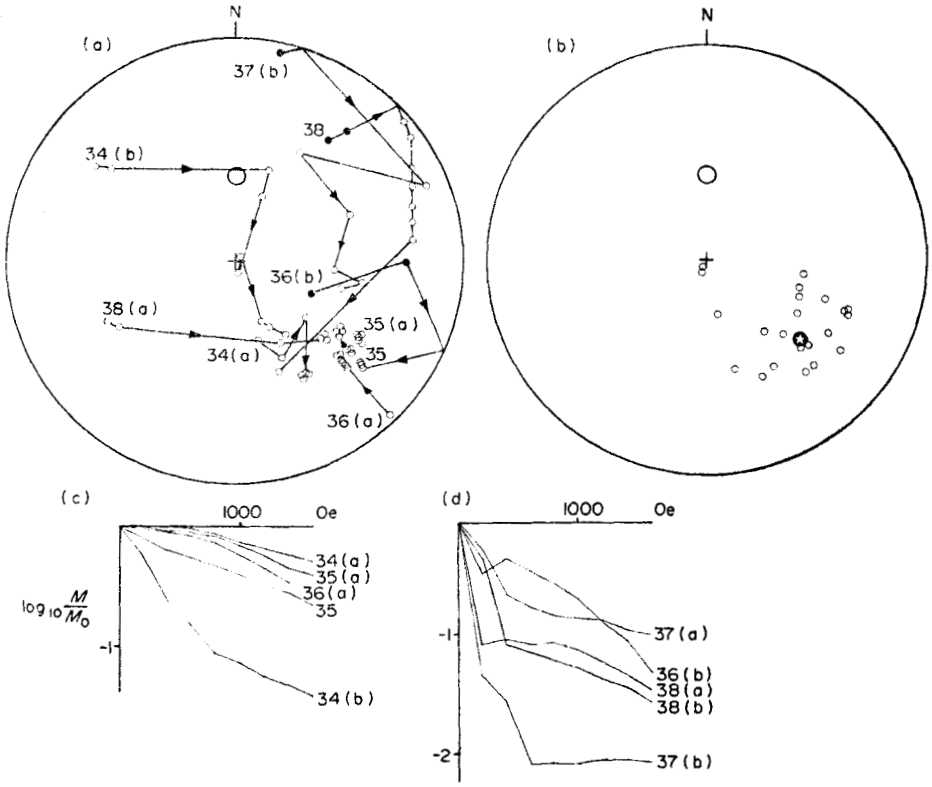


FIG. 5. (a) Behaviour of the pilot samples from the O'okiep basic intrusions with progressive af demagnetization in steps of 100 Oe, (b) cleaned directions of all samples, and (c) and (d) demagnetization characteristics of pilot samples.

Total NRMs of two sites give significant grouping with a mean direction of $D = 117{\cdot}8°$, $I = -36{\cdot}9°$ ($k = 1{\cdot}5$). Pilot samples from sites 34, 35 and 36 show a gradual fall in their moments with af demagnetization while pilots from site 37 and 38 show a more dramatic fall in their moments (Fig. 5(a)). In general magnetic hardness is proportional to stability, and those samples showing large falls in moment also give large shifts in direction of magnetization. With progressive demagnetization directions of all pilots move towards a closely grouped mean (Fig. 5(b)) reached after demagnetizing in fields between 300 and 600 Oe, although it is not reached for samples from site 37 until after demagnetization in fields of 1000 Oe. All sites become significant after af cleaning with a mean direction of $D = 131{\cdot}6°$, $I = -48{\cdot}4°$ ($k = 27{\cdot}8$). This can be compared with total NRM directions obtained from previous sites in the Fonteintjie and Kliphoog bodies respectively of $D = 129°$, $I = -41{\cdot}5°$ ($k = 15$), and $D = 133°$, $I = -50°$ ($k = 58$), which Brock & Piper (1972) used to quote a provisional pole for these intrusions.

These intrusions have a composition between monzonite and diorite with sodic plagioclase predominant and subordinate amounts of hornblende, biotite, ilmenite and copper-bearing sulphides. The rocks are entirely fresh except for sites 35 and 36 where the feldspars are slightly sericitized along grain boundaries and cleavages, and site 37 where sericitization is more general. Ore minerals occur as large rounded blebs, often with re-entrants, and were clearly formed at an early stage of crystallization. Magnetite is entirely present as fine needle-like exsolution lamellae which occur

within fresh ilmenite grains. All sections examined have exsolution lamellae over more than 50 per cent of their area and the highly stable magnetization of these rocks probably results from this exsolution (Strangway, Larson & Goldstein 1968; Evans, McElhinny & Gifford 1968).

3.4 *Lavas of the Sinclair Group, Barby Formation, South West Africa*

Basic lavas of the Sinclair Group (Barby Formation) outcrop over a large area to the north-west of Helmeringshausen and have been folded to acquire dips between 25° and 90° mostly to the east or north. They are predominantly feldspar-phyric varying between andesite and basalt in composition and are associated with subordinate acid lavas and tuffs. Deuteric alteration is practically ubiquitous and volcanism was evidently discontinuous since interbasaltic horizons are common. The sampling sites span most of the thickness of the outcrop and although sites 69–80 form a general stratigraphic sequence (Fig. 6) it is not possible to erect a simple stratigraphy because of extensive faulting (von Brunn 1969); similarly it is impossible to make a precise estimate of thickness of the Barby series. According to B. R. Watters (1973, private communication) the Barby Formation is at least 3000 m and probably as much as 5000 m thick.

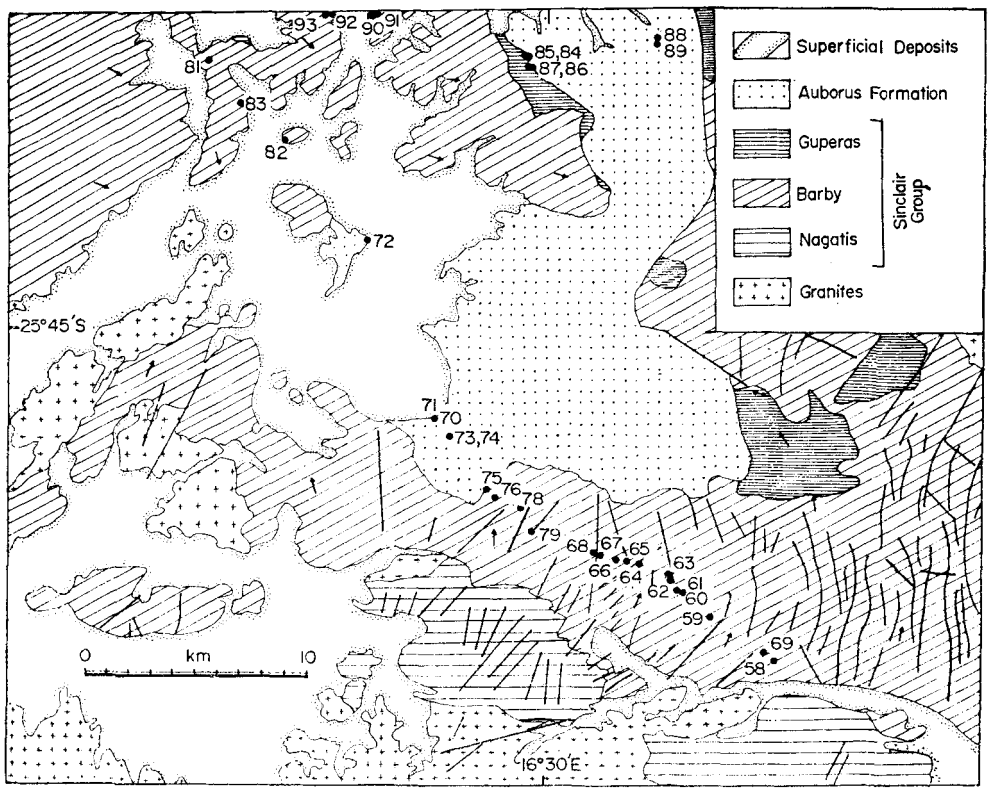


FIG. 6. Location of sampling sites in the region north-west of Helmeringshausen in South West Africa. The geology south of Latitude 25° 45′ S is based on von von Brunn (1969) and north of this on preliminary mapping by B. R. Watters (private communications 1971 and 1973). The location of this region is shown in Fig. 1. Solid lines are major faults and arrows are directions of dip.

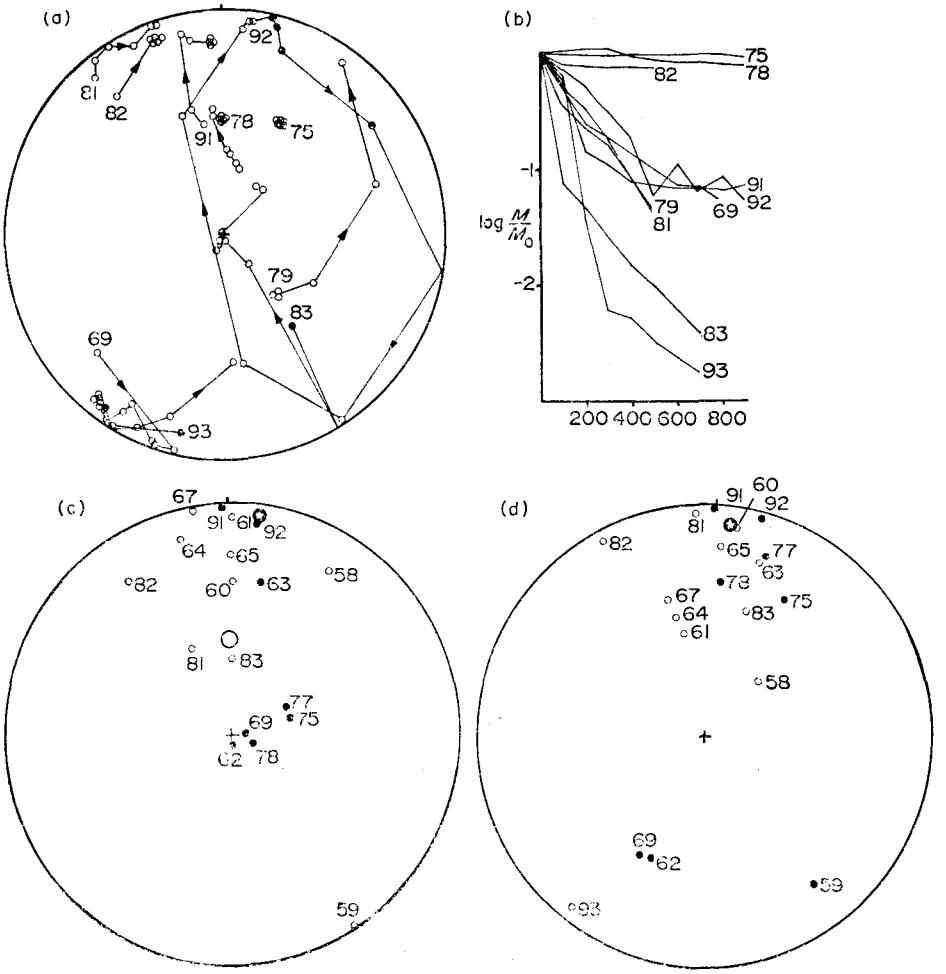


FIG. 7. (a) Directional behaviour of some pilot samples from the lavas of the Barby Formation of the Sinclair Group, South West Africa with progressive af demagnetization in steps of 100 Oe; (b) demagnetization characteristics of these samples; and (c) cleaned site mean directions of magnetization of the Barby group sites before tilt correction. In this and following diagrams the open large circle is the direction of the present Earth's field and the star is the combined mean direction of magnetization.

Intensities of magnetization of these lavas show a wide range of values between 3×10^{-7} to 3×10^{-2} G, with a mean value of 7×10^{-5} G although the distribution of intensities is not geometric and shows no well-defined mean. Behaviour of pilot samples during af demagnetization is correspondingly variable. Several samples, notably the pilots from sites 75, 78 and 82 show practically no decrease in their moments on demagnetizing to 1000 Oe (Fig. 7(a)) and directions of magnetization do not move. Other pilots show moderate (e.g. pilots of sites 69, 79, 81, 91 and 92) to sharp (e.g. pilots of sites 83 and 93) falls in their moments on demagnetizing in fields of a few hundred Oe followed by a more steady, nearly-exponential, fall (Fig. 7(a)). All these samples are however stable, and show little change in directions of magnetization after cleaning in fields of 200–300 Oe. Stability indices reach optimum values mostly between 300 and 500 Oe and continuing demagnetization to 1000 Oe does not

generally improve within-site grouping. The only systematic feature of direction changes during demagnetization is a tendency for some samples to move away from the direction of the present Earth's field, and grouping is improved by af cleaning for all but five of the sites. Sites 82 and 93 which are statistically non-significant before cleaning become so after cleaning but site 79 becomes random at the 95 per cent confidence level (Watson 1956; Irving 1964) after cleaning. Sites 66, 68 and 76 are not significant before or after cleaning.

When the af cleaned site mean directions are corrected for the local dip of the lavas grouping is considerably improved (Fig. 7, Table 3) and the precision k rises from 2·0 to 4·8 significant at the 95 per cent confidence level using the fold test of McElhinny (1964). Scatter is still larger than would be expected from palaeosecular variation at this palaeolatitude (Irving 1964, page 146) and might be improved if it were possible to correct for plunge components. Furthermore most of the sites are situated within small fault blocks and differential rotation of these blocks may contribute to between-site scatter.

Five sites (59, 62, 69 and 93) are magnetized in the opposite sense to the remainder (Table 3) and the first four imply that there are at least two approximate reversals in the lower part of the Barby Series (Fig. 3). Sites 81, 82, 83, 91 and 93 are from opposite sides of a synclinal structure and the single reversal represented by site 93 may represent the same horizon as 59, 62 and 69, although the lava of site 93 is petrographically distinct, being markedly feldspar phyric.

Table 3

Lavas of Barby Formation, Sinclair Group, South West Africa

			After cleaning, corrected for dip						
Site No.	Cleaning field (Oe)	No. of samples	D	I	R	k	α_{95}	Pole position °E	°N
58	300	5	42	−63	4·7	14·0	21	146	52
59	400	6	140	+23	6·0	255·0	4	299	50
60	250	6	6	−9	4·9	4·7	35	32	68
61	250	6	344	−52	2·8	9·9	42	257	75
62	250	6	203	+43	5·4	8·0	25	42	35
63	400	5	14	−22	4·3	6·1	34	62	70
64	300	5	344	−45	4·2	4·9	39	279	75
65	300	4	2	−19	3·9	30·1	39	23	74
67	400	5	341	−38	4·1	4·5	41	298	72
69	300	5	208	+42	3·7	3·0	53	48	33
75	350	6	27	+32	6·0	179·7	5	50	40
77	350	5	16	+21	4·9	32·1	14	42	50
78	350	5	3	+34	4·9	72·2	9	20	46
81	300	6	355	−2	5·9	38·4	11	4	65
82	300	4	329	−4	3·8	117·3	23	321	52
83	400	4	16	−42	3·8	150·5	24	104	75
91	500	5	359	+2	4·8	20·3	17	14	63
92	350	6	11	+4	5·7	19·6	16	40	60
93	350	7	217	−9	6·2	7·3	24	72	43
Group mean statistics after dip correction with unit weight to each sample (i) and each site (ii)									
	(i)	98	5·3	−4·4	72·6	3·8	8·5	29·7	62·9
							($d\Psi = 4·4$, $d\chi = 8·5$)		
	(ii)	19	4·4	−7·4	15·2	4·8	17·3	28·0	67·5
							($d\Psi = 8·7$, $d\chi = 17·4$)		
Group mean statistics before dip correction with unit weight to each sample (i) and each site (ii)									
	(i)	98	11·1	−5·6	42·9	1·8	16·1	38·7	59·5
	(ii)	19	9·0	5·4	9·9	2·0	33·3	34·8	60·2

Petrology of these lavas has been treated in detail by von Brunn (1969) and only relationship between petrology and magnetic properties is considered here. They are mostly slightly undersaturated and near the junction between andesite and basalt in composition; the bulk are zeolitized. The state of alteration varies from greeschist–amphibolite facies metamorphism (e.g. sites 60–65) where the feldspar and pyroxenes are pseudomorphed by sericite, and chlorite and actinolite respectively, to zeolite–greenschist (e.g. sites 82, 83) in which a primary igneous texture is visible but the pyroxenes and olivines are pseudomorphed, to zeolite facies (e.g. sites 90–93) in which the igneous minerals are entirely fresh. Thus in general degree of alteration diminishes northwards across the sampling area (Fig. 6), and has affected the magnetic minerals to varying degrees. Unoxidized titanomagnetite grains in the more highly altered rocks are surrounded, and sometimes completely pseudomorphed, by alteration rims of rutile and haematite; grains also show variable amounts of granulation (Ade-Hall *et al.* 1968, Part I) within their area. Lavas with an intermediate degree of alteration are at the most marginally granulated and titanomagnetites, generally with oxidation states 1–2, are preserved with slight marginal alteration. Sites 90–93 show an absence of granulation and subdivision by ilmenite lamellae equivalent to oxidation states 2–3, and locally, a pesudomorphing of the ilmenite lamellae gives oxidation states 4 and 5. Site 92 shows two stages of titanomagnetite growth with large, partially resorbed, grains and fine elongated grains in the groundmass and some other sections (e.g. 91 and 93) have fine-grained magnetite pseudomorphing olivine. Although the Barby Formation lavas exhibit degrees of low-grade alteration the magnetic evidence suggests that remagnetization has not been widespread. Firstly the positive fold test demonstrates that the magnetization of these lavas predates that of the Guperas lavas (post folding) and secondly reversals (sites 59 and 62) are found even in the highly altered rocks. Any remagnetization therefore must have been at least pre-folding and local in its effects, and the observation of Ade-Hall *et al.* (1968, Part II) that titanomagnetite granulation does not apparently produce remagnetization seems to receive some support here.

3.5 *Lavas of Sinclair Group, Guperas Formation, South West Africa*

Sites 84, 85, 86 and 87 were taken through a thin suite of strongly oxidised basalt lavas at 16·5° E, 25·6° S. They comprise several flow units but no sedimentary horizons and may represent one eruptive episode only. Samples from the Guperas Formation possess magnetic moments which diminish only gradually with increasing demagnetizing field (Fig. 8(a)) and possess high stability (Fig. 8(b)). Grouping of all sites is good at the total NRM level and they combine to give a mean direction $D = 334°$ and $I = -23°$. After demagnetizing in fields of 300 Oe overall precision

Table 4

Lavas of Guperas Formation, Sinclair Group, South West Africa

		Cleaned directions, site mean statistics							
Site No.	Cleaning field (Oe)	N	D	I	R	k	α_{95}	Pole position °E	°N
84	300	5	337	−20	4·0	37	12	318	64
85	300	4	337	−20	3·9	28	17	318	64
86	300	4	338	−18	4·0	342	5	321	64
87	300	3	332	−21	3·0	140	10	311	60
		Group mean statistics, cleaning directions							
		4	336·4	−19·9	4·0	871·0	3·1	316·6	62·9
							($d\Psi = 1·7$, $d\chi = 3·3$)		

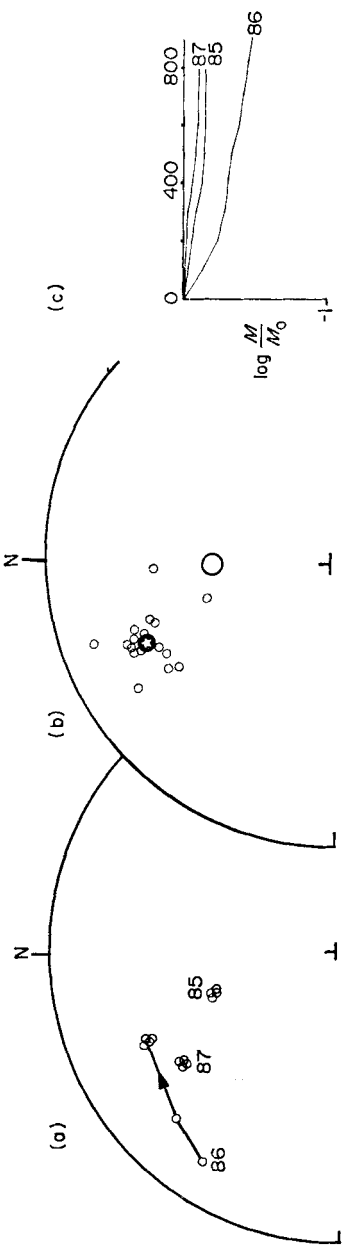


FIG. 8. (a) Behaviour of pilot samples of the Guperas Formation of the Sinclair Group with stepwise af demagnetization; (b) cleaned directions of magnetization of all samples; and (c) demagnetization characteristics of the pilot samples.

improves from 156 to 871 but the mean direction is only slightly altered to $D = 336°$ and $I = -20$. They are magnetized in the same sense as the bulk of the Barby Formation.

These are basalt lavas with vesicles infilled with carbonate and zeolites. Plagioclase phenocrysts are slightly sericitized with primary mafic minerals entirely converted to secondary minerals. Large grains of haematite occur as late-stage magmatic minerals enclosing feldspar laths and are themselves marginally overgrown by fine-grained haematite. Haematite is also a prominent constituent of the groundmass where grain size goes down to the limit of microscopic resolution; a few relict titanomagnetite grains have also been observed in a sample from site 86. Predominance of haematite as an opaque phase in these lavas explains the high magnetic coercivity and stability, and the petrographic relationships suggest that it developed under strongly oxidizing conditions at a late magmatic–early deuteric phase.

3.6 *Sediments of the Auborus formation, South West Africa*

Fine red sandstone and siltstone members of the Auborus formation were samples at different horizons in three localities. Sites 70, 71, 72 and 73 were sampled in the vicinity of 16·4° E, 25·7° S, 74 at 16·5° E, 25·8° S, and sites 88 and 89 at 16·5°E, 25·6° S. Total NRMs of 36 samples from eight sites in the Auborus formation form a compact grouping with a southern declination and a negative inclination (Fig. 9)

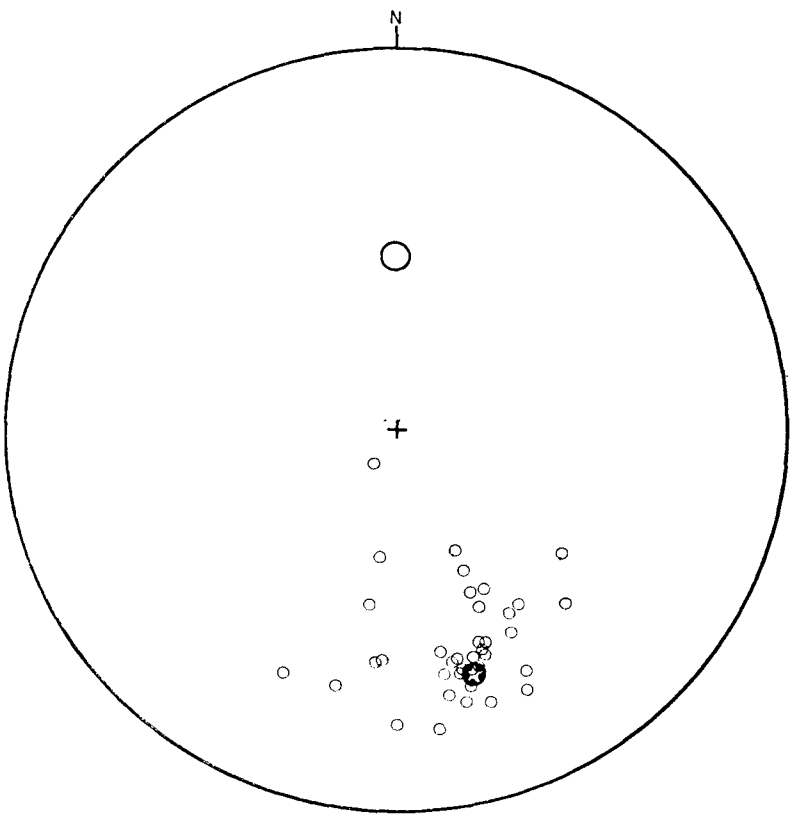


FIG. 9. Total NRM directions of magnetization of samples from the Auborus Formation, South West Africa.

Table 5

Sediments of the Auborus Formation, South West Africa

Site No.	NRM site mean directions						Palaeomagnetic pole	
	N	D	I	R	k	α_{95}	°E	°S
70	4	164	−22	3·9	411·4	4	172	50
71	4	143	−44	3·6	8·2	34	158	27
72	5	193	−24	4·9	72·3	9	216	50
73	6	165	−24	5·9	132·2	5	174	49
74	6	165	−27	5·9	224·7	4	175	47
88	5	152	−49	4·8	22·7	16	169	28
89	4	152	−27	3·0	52·6	12	159	42
90	3	164	−41*d*	2·9	25·1	25	178	38
	8	163·3	−33·1	7·7*d*	24·1	11·5	174·3	43·3
						($d\Psi = 7{\cdot}4$, $d\chi = 13{\cdot}1$)		

well removed from the present Earth's field. The eight sites combine to give a mean direction of $D = 163°$, $I = -33$ ($k = 24$) and are thus magnetized in opposite sense to the Guperas Formation lavas and the bulk of the Barby Formation. This collection has not yet been thermally cleaned but systematic magnetic cleaning of several pilot samples in fields up to 1600 Oe demonstrates that magnetizations of these samples (Fig. 14) have high stability to af fields.

3.7 *Basic dykes, Klein Karas, South West Africa*

Sites 104, 105, 106 and 107 are taken from dykes that outcrop in the vicinity of Klein Karas. These dykes have a north–south trend and cut granite gneisses in this region. Two K–Ar age determinations have been made on one of the dykes from 17° E to 28° S, of 878±41 My and 543±15 My, respectively (Dr N. J. Snelling 1973, private communication). De Villiers (1968, p. 35) interpreted the higher figure as a minimum age and a possible association of these dykes with the Richtersveld Igneous Complex (850 My) suggests itself. Recent work has shown that at least several of these dykes may cut a late granite of the Richtersveld suite, dated at 720 My (Welke *et al.* 1973) but whether these include the sampled dykes is not known.

Samples have intensities between $2{\cdot}0\times10^{-4}$ and $2{\cdot}0\times10^{-2}$ with a mean of $9{\cdot}5\times10^{-4}$ G, and typically show gradual fall in their moments on af demagnetization (Fig. 10(a)) with slight to moderate changes in direction (Fig. 10(b)). Grouping of all sites is improved by af demagnetization with optimum stability indices reached after demagnetizing in fields of 200–400 Oe. Sites 104, and 106 which are not statistically

Table 6

Dykes, Klein Karas, South West Africa

Site No.	Clean directions, site mean statistics						Pole position	
	N	D	I	R	k	α_{95}	°E	°N
104	3	287	−15	3·0	59·8	16	289	19
105	4	251	2	4·0	65·4	11	280	−17
106	3	314	−11	2·9	22·2	27	306	41
107	2	308	−1	2·0	47·7	37	307	33
Group mean statistics after cleaning:								
(i)	12	285	−7	10·5	7·4	17	292	15
(ii)	4	290	−7	3·6	5·1	35	294	20

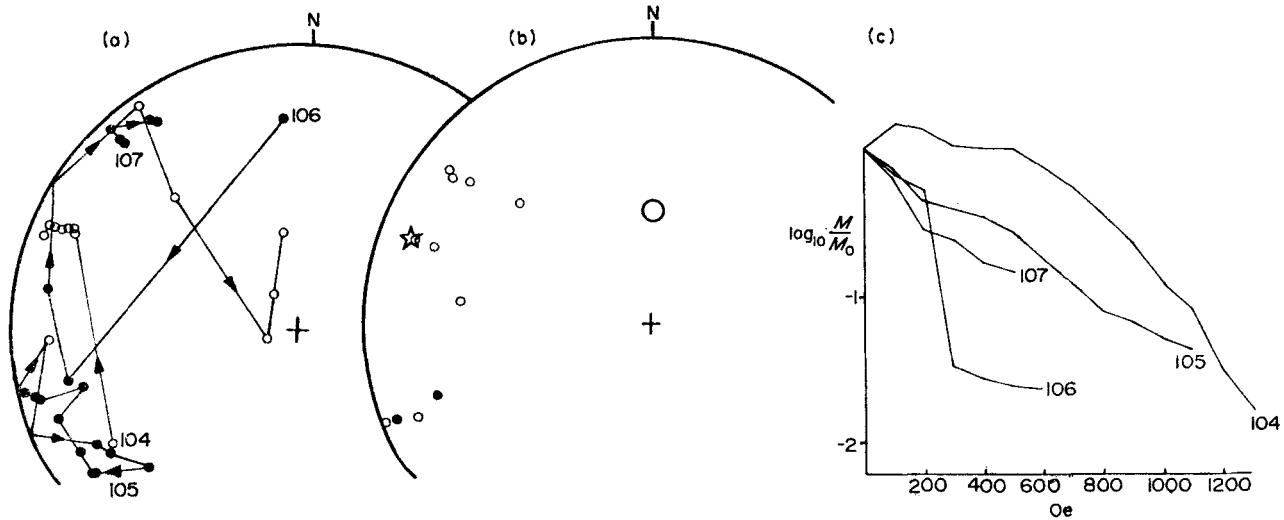


FIG. 10. (a) Directional behaviour of pilot samples from basic dykes near Klein Karas, South West Africa with af demagnetization in steps of 100 Oe; (b) sample cleaned directions; and (c) demagnetization characteristics of the pilot samples.

significant before cleaning become so after cleaning and the sites combine to give a mean direction of $D = 290°$, $I = -7°$ ($k = 5{\cdot}1$). Titanomagnetite is present in these dykes intersertal to the feldspars and has been subjected to extensive deuteric alteration with dense exsolution of ilmenite lamellae. The magnetite remains fresh, the ilmenite lamellae are variably altered to aggregates of rutile and haematite (oxidation class 4–5). A general correlation of magnetic stability with oxidation state has been noted elsewhere (Watkins & Haggerty 1965; Ade-Hall *et al.* 1968, Part II) but the high oxidation state observed here is exceptional for dyke material (Ade-Hall & Lawley 1969).

3.8 *Red sediments, Doornpoort Formation, South West Africa*

Massive purple siltstones belonging to the Tsumis series were sampled in two regions approximately 60 km north of Kalkrand. Sites 52 and 53 were taken near 17·2° E, 23·7° S where the beds dip 18° in a direction 250° E and sites 54, 56 and 57 were taken 2–3 km further south where they dip 45° in a direction 330° E. These localities have a marked fabric and cleavage is well developed in the finer-grained horizons.

The samples have NRM intensities between 3×10^{-6} and 5×10^{-5} G with a mean of $1{\cdot}5 \times 10^{-5}$ G, giving directions with a south-westerly declination and negative inclination. Samples exhibit only slight changes in direction when treated in alternating fields to 1600 Oe with no significant fall in moments. Grouping is significantly better at the 95 per cent confidence level (McElhinny 1964) *before* correcting for the dip of the sediments and the precision deteriorates from $k = 58{\cdot}3$ to $k = 8{\cdot}1$ (total NRMs) when the site mean directions are corrected for dip (Fig. 11, Table 7). This formation is strongly cleaved and has clearly undergone remagnetization with the imposition of the metamorphic fabric. The combined mean direction uncorrelated for dip is $D = 229°$, $I = -43°$ and significantly removed from the present Earth's field.

3.9 *Pre-Nama dykes, Namaqualand, Cape Province*

Dolerite dykes post dating the last metamorphic events of the Namaqualand granite gneiss massif intrude the granites in a broad region that extends northwards from Nababeep to the Orange River. They have a north to NNE trend and are overlain unconformably by sediments belonging to the Nama system. Six of these dykes were sampled at three localities: 17·5° E, 29·6° S (sites 40, 41), 17·9° E, 29·1° S (sites 42, 43), and 17·6° E, 28·8° S (sites 44 and 45).

Table 7

Sediments of the Doornpoort Formation, South West Africa
NRM site mean directions before dip correction

Site No.	N	D	I	R	k	α_{95}	Palaeomagnetic pole °E	°S
52	5	231	−48	5·0	84·2	8	242	17
53	5	226	−52	4·8	29·2	15	236	18
54	5	233	−35	4·8	25·0	15	252	23
55	3	230	−56	2·9	14·6	33	236	13
56	5	223	−33	5·0	14·5	6	246	30
57	5	232	−34	4·9	77·0	8	252	24
	6	229·3	−43·0	5·9	58·3	8·8	224·9	21·8
							($d\Psi = 6{\cdot}8°$, $d\chi = 11{\cdot}0$)	
Group mean direction after dip correction:								
	6	212·2	−24·7	5·4	8·1	25·1	241·3	41·7
							($d\Psi = 14{\cdot}4$, $d\chi = 26{\cdot}9$)	

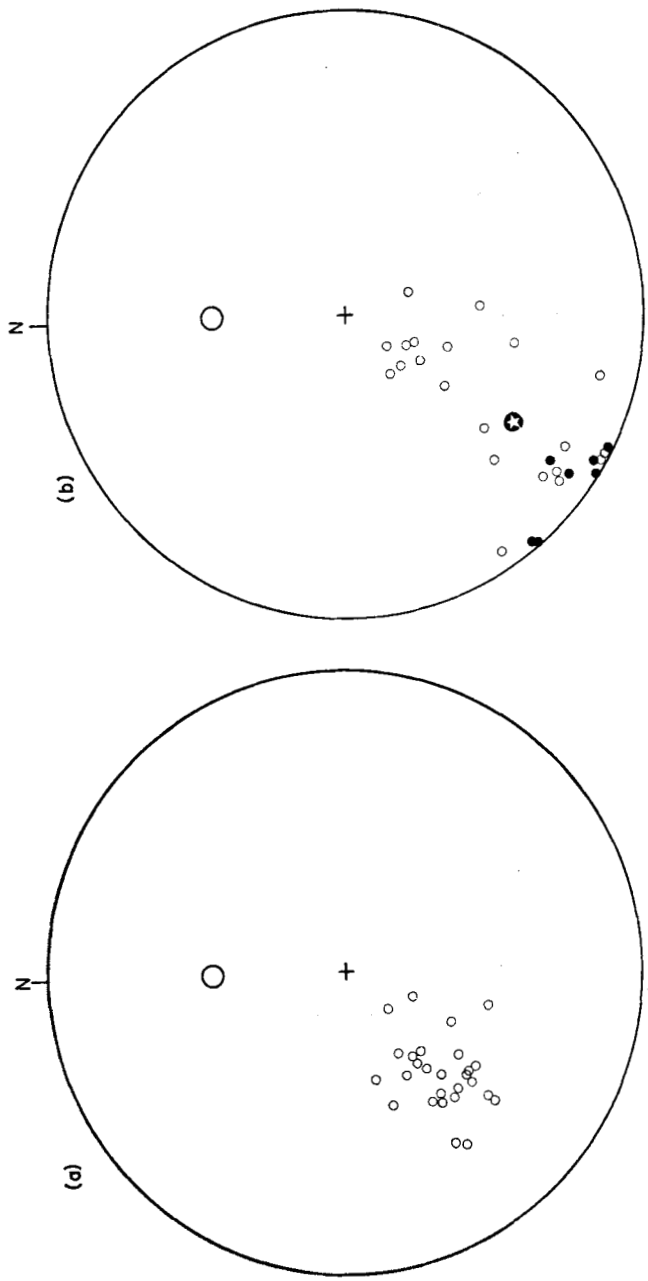


FIG. 11. NRM directions of samples from the Doornpoort Formation, South West Africa, (a) before; and (b) after tilt correction

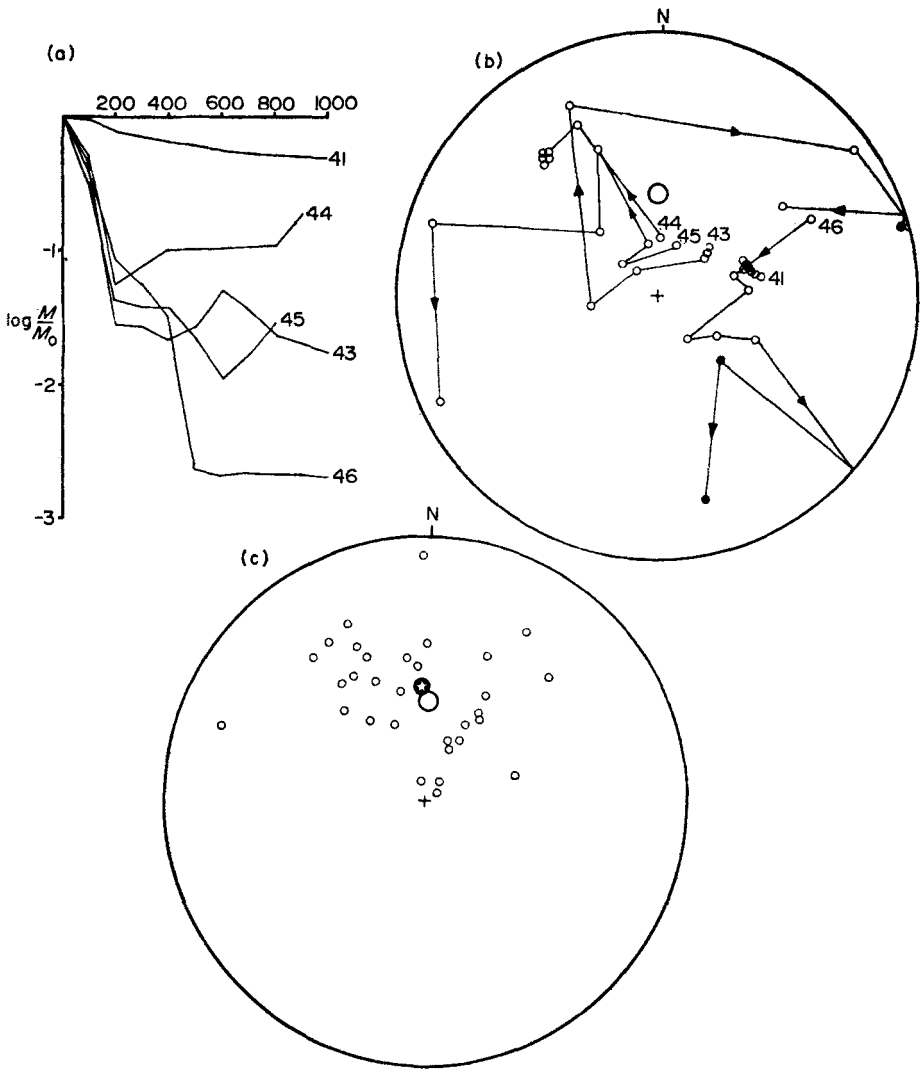


FIG. 12. (a) Demagnetization characteristics; and (b) directional behaviour of pilot samples from dolerite dykes, Namaqualand, Cape Province, with progressive demagnetization in steps of 100 Oe; and (c) cleaned directions of magnetization of samples from these dykes.

The dyke samples have intensities of magnetization varying between 3×10^{-5} and 4×10^{-3} G with a geometric mean value of $4{\cdot}5 \times 10^{-4}$ G. The NRM grouping of all sites is statistically significant at the 95 per cent confidence level and the six sites combine to give a mean direction of $D = 1°$, $I = -66°$ ($k = 14{\cdot}4$) which is close to, but significantly removed from, the direction of the present Earth's field. Pilot samples exhibit moderate to low stabilities on af demagnetization: the pilot from site 41 shows only a slight and exponential fall in moment with stepwise treatment, with slight change in direction. All other pilots show sharp fall in moments after treatment in fields of 200 Oe (Fig. 12(a)) followed by erratic behaviour related in no way to movement of the magnetic vector. Sites 40 and 43 for example show little movement of the vector for the first few hundred Oersteds of treatment followed by

Table 8

Pre-Nama dykes, Namaqualand, Cape Province
Cleaned directions, site mean statistics

Site No.	Cleaning field (Oe)	No. of samples	D	I	R	k	α_{95}	Palaeomagnetic pole position °E	°N
40	300	6	23·9	−33·3	4·5	3·8	39	85	65
41	300	7	358·3	−60·9	6·4	9·9	20	203	78
42	600	5	336·8	−44·6	4·9	4·9	11	291	69
44	100	5	327·8	−45·1	4·9	39·2	12	285	62
45	600	5	31·5	−68·9	4·3	5·8	34	16	58
			Group mean statistics						
	(i)	28	356·3	−53·4	23·8	6·4	11·7	232·5	84·6
	(ii)	5	357·1	−53·3	4·7	13·3	21·7	227·5	85·1

($d\Psi = 20{\cdot}9$, $d\chi = 30{\cdot}2$)

larger changes beyond this, but site 44 exhibits little change in direction after the first 200 Oe although the moment begins to increase, while the pilot for site 45 shows greater stability between 200 and 500 Oe. In general precision deteriorates with af demagnetization and within-site grouping is only improved for site 41 while site 43 becomes insignificant. Samples retain a predominant northerly declination and negative inclination however (Fig. 12(b)) and the overall mean direction after cleaning is $D = 357°$, $I = -53°$.

These dykes show variable degrees of alteration from slight sericitization of the feldspars (sites 41 and 42) and marginal chloritization of the pyroxenes (43) to practically complete pseudomorphing of silicates by secondary minerals (45). Opaques present in these rocks include pyrite and large, generally unoxidized, titanomagnetites intersertal with respect to the silicates. These titanomagnetites have irregular, probably resorbed, outlines and represent an early magmatic phase, and at site 45 they are largely replaced by secondary aggregates. Titanomagnetite also occurs in all sections as fine acicular grains in the groundmass which appear to be largely unoxidised. The low magnetic stabilities of this collection are probably attributable to the unoxidized state of the bulk of the titanomagnetite.

3.10 *Fish River Formation, Nama Group, South West Africa*

This series was sampled in a tributary canyon to the Fish River Canyon between 10 and 15 km west of Seeheim (17·7° E, 26·8° S). The samples are entirely medium to fine-grained sandstones dipping uniformly at 10° in a direction 120° E and the formation has a strong red colouration throughout.

Intensities of magnetization vary between 8×10^{-7} and 5×10^{-5} G with a mean of $3{\cdot}6 \times 10^{-6}$ G. NRMs give directions with a northerly to easterly trend and negative inclination. This is close to the direction of the present Earth's field but presence of a primary component is suggested by the fact that correction for the regional dip moves the overall mean from $D = 27$, $I = -56$ to $D = 13$, $I = -54$ or towards the present field direction. One or two samples from each site have been subjected to af demagnetization in steps of 200–1600 Oe in the course of which they characteristically lose between 20 and 50 per cent of their moment and undergo slight to moderate changes in direction (Fig. 14(a) and (b)). This behaviour implies that the magnetization, at least in part, is held by magnetite and contrasts with that of the red sediments of the Auborus Formation and Doornpoort Formation which undergo practically no change in direction or magnitude of their moments with af demagnetization, implying a magnetization held by members of the ilmenite-haematite

Table 9

Fish River Series, Nama system, South West Africa

Site No.	N	D	I	R	k	α_{95}	Palaeomagnetic pole °E	Palaeomagnetic pole °N	Cleaning field (Oe)	N	D	I	R	k	α_{95}	Palaeomagnetic pole °E	Palaeomagnetic pole °N
94	5	12	−31	4·7	15·2	20	67	75	1000	5	2	−16	5·0	11·2	7	23	71
95	5	9	−50	4·8	16·2	20	139	81	1000	5	8	−54	4·6	10·7	24	151	80
96	5	30	−46	4·9	39·0	12	115	63	1000	6	37	−52	5·9	34·2	12	126	57
97	5	346	−47	4·9	33·9	13	280	78	1000	5	350	−37	4·7	13·7	21	319	79
98	6	16	−55	4·8	4·1	38	152	77	1000	4	101	−51	3·8	16·0	24	140	5
99	5	55	−67	4·8	20·1	26	152	42	1000	6	91	−52	5·8	28·6	13	139	13
100	3	5	−54	2·9	23·9	29	170	81	1000	3	2	−49	2·9	14·8	33	168	86
101	5	26	−60	4·5	7·9	45	148	64	1000	6	91	−56	4·8	4·1	38	142	15
102	4	352	−65	3·4	4·8	10	213	69	1000	4	125	−51	3·9	21·9	20	152	−11
Group mean statistics, before af cleaning:									Group mean statistics, after af cleaning and dip corrections:								
(i) dip corrected									(i) all samples								
	9	13	−54	8·7	26·5	10	147	77		44	42	−57	34·8	4·7	11	136	53
(ii) before correction									(ii) all sites								
	—	27	−56	—	—	—	138	65		9	42	−58	7·5	5·6	23	137	53

($d\Psi = 25{\cdot}4$, $d\chi = 34{\cdot}4$)

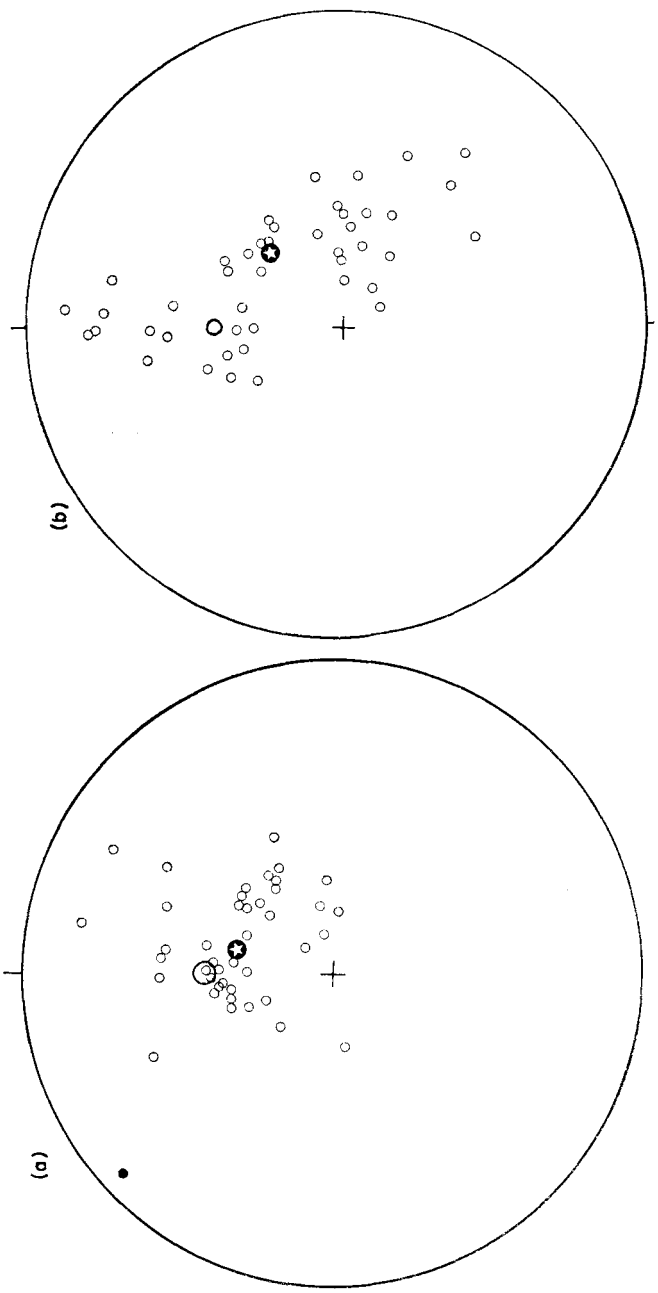


FIG. 13. Tilt corrected directions of magnetization of samples from the Fish River Formation of the Nama Group (a) before; and (b) after af demagnetization.

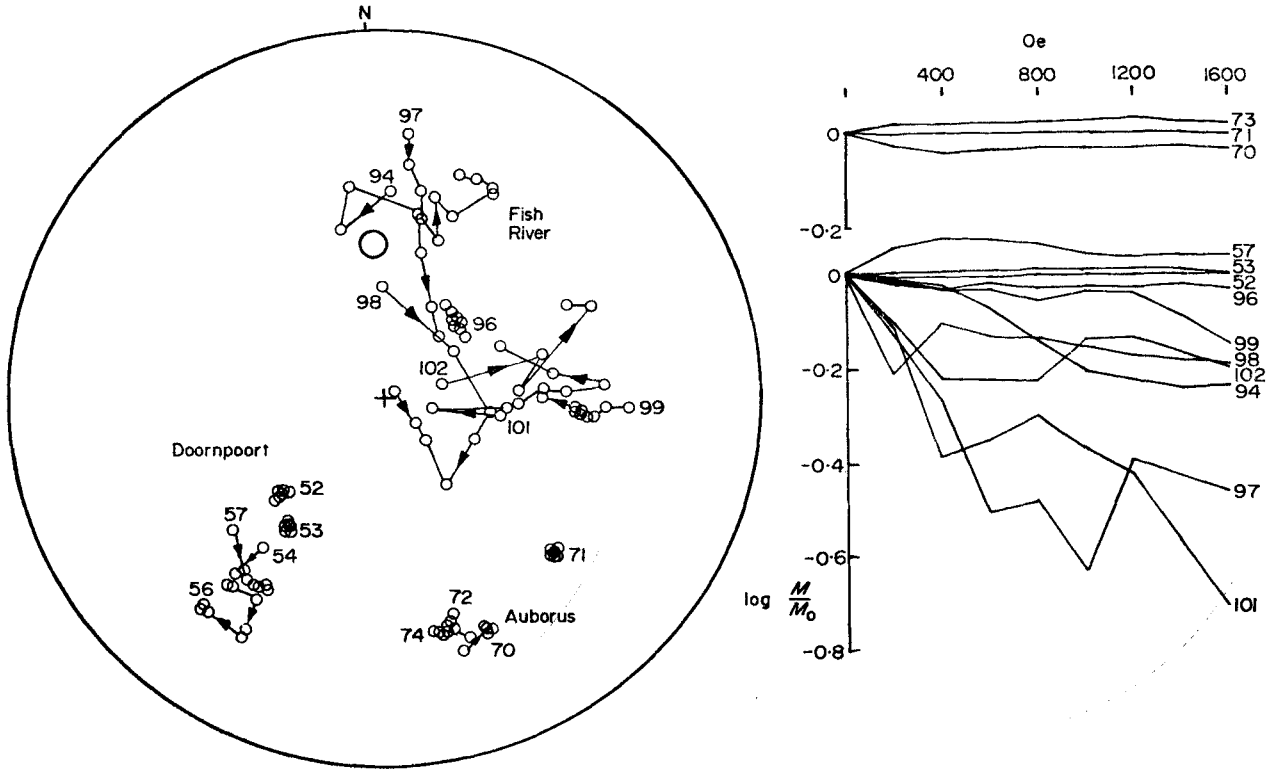


FIG. 14. Behaviour of (a) directions; and (b) moments of magnetization of sediment samples from the Auborus Formation (sites 70, 71, 72 and 74), the Doornpoort Formation (sites 52, 53, 54, 56 and 57) and the Fish River Formation (sites 94, 96, 97, 98, 99, 101 and 102) with progressive af demagnetization in steps of 200–1600 Oe.

Series (Nagata 1967). In several cases (e.g. pilots from 97, 98 and 101, Fig. 14) the direction of NRM moves systematically away from the present Earth's field and af demagnetization appears to be successful in removing a secondary component near this direction. To test this possibility the entire collection has been demagnetized in fields of 500–600 Oe and 1000 Oe, and the overall grouping is slightly better after demagnetizing in the higher field than in the lower field, although the combined precision ($k = 5{\cdot}6$) is considerably lower than that of the total NRM grouping ($k = 26{\cdot}5$). Sites 98, 99, 101 and 102 move away from the present Earth's field direction and three of them improve their precisions with cleaning. Remaining sites stay close to their total NRM directions but their precisions deteriorate and in view of the possibility that the combined mean of Table 9 includes sites which have not been successfully cleaned by the af method, poles for individual sites are quoted. The distribution of mean directions from this formation is not Fisherian and an explanation for this is explored in the next section.

4. Interpretation of virtual geomagnetic poles

4.1 *Orange River Lavas, Cape Province*

Since these lavas have been subjected to at least two later thermal events, age dating is likely to prove difficult, but is currently being attempted (D. L. Reid 1972, private communication). With a possible age between *c.*2600 and 1850 My the pole merits comparison with the apw curve for Africa during this interval (McElhinny *et al.* 1968; Piper, Briden & Lomax 1973) and falls close to the curve at *c.*2100 My although the antipole also falls near the curve at *c.*2500 My and the lavas might possibly be as old as this.

4.2 *Upper Proterozoic results*

The only published pole of similar age to the oldest Upper Proterozoic pole of this study is that obtained by Gough (1956) from five sites in the Pilanesberg dykes. An age for this pole of 1310 ± 60 My is suggested by Rb–Sr determinations although an older age, possibly near 1420 ± 70 My is suggested by associated rocks (see Clifford 1968, page 345). An apparent movement of the pole is implied across central Africa between the times of intrusion of these dykes and the time of formation of the O'okiep intrusions (pole 3, Fig. 15) and the eruption of the Kisii series of Kenya (pole 2, Fig. 15). Further dating of the Kisii Series by the $^{40}Ar/^{39}Ar$ technique (Charlton 1973) has suggested an age of approximately 1200 My for this pole.

Relative movement of the pole away from Africa through at least 65° is implied by the pole for the Barby Formation (pole 5, Fig. 15) with a maximum age of 1290 ± 80 My and a minimum age of 1020 ± 70 My. A single age of 500 ± 20 My from the author's site 75 (D. C. Rex 1973, private communication) is most probably to be interpreted in terms of argon loss during the ' Pan African ' thermal-tectonic episode. This pole is not significantly different from the poles obtained from the Umkondo and Waterberg lavas and intrusions of Rhodesia and the Transvaal (poles 4, Fig. 15) and this palaeomagnetic evidence is an indicator to the true age of the Umkondo–Waterberg igneous activity. Johnson & Vail (1965) report a single age of 1140 My for a dolerite cutting the Umkondo but Jones & McElhinny (1966) report inconsistent K–Ar ages ranging from 600–1300 My for Waterberg dolerites and 650–1150 My for Umkondo dolerites. If the problem with this material is predominantly argon loss we may suspect that the higher ages are near the true age of this material. The correlation with the pole from the Barby series defines a minimum age of 1020 ± 70 My for this igneous activity. The Waterberg and Umkondo systems were originally thought to be pre-Upper Proterozoic in age (Johnson & Vail 1965) and this

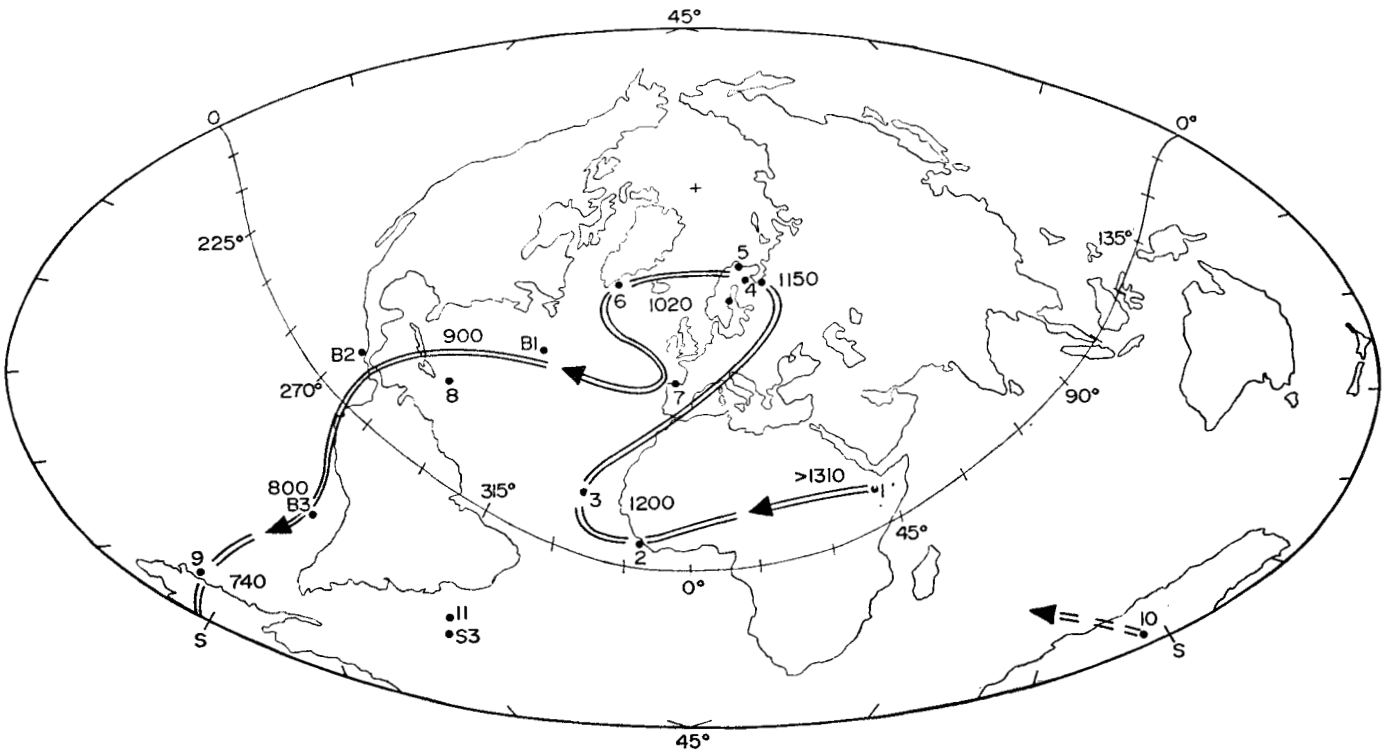


FIG. 15. Apparent polar wander path for Africa from *c.*1300 to *c.*650 My plotted on an oblique equal area projection. The poles are numbered as they are referred to in the text. They are: 1, Pilanesberg dykes, South Africa, > 1310 My; 2, Kisii Series, Kenya, *c.*1200 My; 3, O'okiep intrusions, South Africa, *c.*1200 My; 4, Waterberg and Umkondo volcanics of Botswana, Rhodesia and South Africa; 5, Barby Formation, South West Africa, 1200–1020 My; 6, Guperas Formation, South West Africa, < 1020 My; 7, Auborus Formation, South West Africa; B1, Bukoba Sandstone, Tanzania, *c.*1000 My; 8, Klein Karas dykes, South West Africa, 878 My; B2, Kigonero Flags, Tanzania, > 890 My; B3, Gagwe Amygdaloidal Lavas, Tanzania, 810 My; 9, Mbozi Complex, Tanzania, 743 My; 10, Pre-Nama dykes, South Africa, 653 My; 11, Fish River Series, South West Africa, Lower Cambrian; S3, ' soft ' magnetization, Orange River Lavas, South Africa. Pole 11 and probably pole S3 belong to a later portion of the apw curve than that depicted here.

conclusion still applies to the Waterberg system which is only *intruded* by these dolerites, and may be older than 1720 My (Jones & McElhinny 1967). However, the pole obtained from Umkondo *lavas* places a likely age on the Umkondo and makes a correlation of this System with part of the Upper Proterozoic of central Africa (Swift 1961; Clifford 1967, page 45) again a probability.

The poles 6 and 7 from the Guperas lavas and Auborus Formation respectively define by their superposed relationship with the Barby Formation a return movement of the pole towards Africa. They both postdate the Rooikam granite (von Brunn 1969) and hence the movement from pole 5 to pole 6 can be dated as taking place at sometime around 1020 ± 80 My. The loop B1-B2-B3 of Fig. 16 has been defined (Piper 1972) from study of the Bukoban system of Tanzania, the Bukoba Sandstone pole B1 has a probable minimum age of 990–1000 My and the pole B2 from the Kigonero Flags has a probable minimum age of 890 My (Dr N. J. Snelling, unpublished results; Professor L. Cahen 1972, 1973, private communications). This polar wander curve derived from the Tanganyika craton can be incorporated in the curve derived from south-western Africa because close agreement has been observed between poles *c.*2300–1950 My in age from cratonic regions of west and southern Africa (Piper *et al.* 1973); the Kisii and O'okiep poles and the Klein Karas (pole 8, Fig. 15) and Kigonero Flags (>890 My) poles are also consistent with a single apparent polar wander curve.

The position of the Auborus pole (7) in relation to the Bukoba Sandstone pole B1 on the polar wander curve suggests that it is slightly older than this formation (>990–1000 My). The palaeomagnetic information precludes it being greatly younger than the Sinclair Formation and this is in agreement with its post-orogenic nature; the palaeomagnetic inclination shows that these haematite-rich sediments were deposited at a palaeolatitude of 18°. Pole S3 is derived from the 'soft' component shown by sites 46, 47 and 49 in the Vioolsdrif lavas is merely inserted here to show that its position is commensurate with acquisition in Late Precambrian times. It is known that this region was subjected to thermal events at this time including the activity of the Richtersveld complex (50 km to the west) and extensive intrusion of dolerite dykes in the immediate vicinity before deposition of the Nama System. There are thus several possible geological causes for remagnetization although no single one can be isolated.

The pole 10 from the pre-Nama dykes of Namaqualand is removed by 63° from the section of the African polar wander curve at B3 dated at *c.*800 My. The pre-Nama dykes west of Vioolsdrif were dated by Köstlin (1971) at 653 ± 70 My and this pole is therefore younger than the pole 9 (Fig. 15) from the Mbozi Complex of Tanzania dates at 743 ± 30 My (Author, unpublished result). There is some doubt as to whether the five dykes used to compute this pole are all in fact pre-Nama in age and Dr S. R. Charlton (1973, private communication) has obtained Karroo ages from an earlier unsuccessful palaeomagnetic collection of dykes in Namaqualand made by Dr C. Snape. A Karroo age would not be inconsistent with the directions of magnetization derived from this material but cannot be substantiated for three of the dykes. In addition to the age data of Köstlin (1971), sites 44 and 45 predate the Nama unconformity and a sample from site 42 has been dated at 401 ± 16 My by the K–Ar method (D. C. Rex, 1973, private communication) an age which is again most reasonably interpreted in terms of thermal overprinting and argon loss. There is no direct evidence bearing on the age of the other two dykes and in view of this the combined pole position should be treated as provisional.

McElhinny, Giddings & Embleton (1974) recognize a large loop in the apparent polar wander curve for Gondwanaland between Late Precambrian and Ordovician times, which implies that the pole moved from a position close to that indicated by the pre-Nama dyke and Mbozi poles to the vicinity of West Africa as suggested by

the pole from the Ntonya ring structure of Malawi (Briden 1968, 630±24 My). The pole then followed a large loop which took Africa to low latitudes in Lower to Middle Cambrian times and back to high latitudes in Ordovician times. The poles derived from the Fish River Formation of this study appear to match the part of the loop proposed by McElhinny *et al.* (1974) close to the portion falling between *c.*600 My and Lower Cambrian times consistent in part with the age inferred from palaeontological data. The polar migration is also consistent with palaeoclimatic indicators which suggest that the climate ameliorated during deposition of the Nama Group with deposition of glacial horizons in the lower part and red sediments (Fish River Formation) in the upper part (Dr A. Kröner 1973, private communication).

The Doornpoort Formation from which pole 12 is derived is separate from the Nama Group by two unconformities and predates the Nosib and Gariep Groups. It is therefore older than 800 My and may be as old as the Auborus Formation. It cannot however be matched to any part of the African apw curve from 1300 to 600 My and since the negative fold test implies a post-tectonic magnetization it probably correlates with folding of the Damaran belt. In Late Precambrian and Lower Palaeozoic times in Africa Clifford (1967) distinguishes a Damaran episode with a main pulse between 500 and 550 My and a slightly earlier Katangan episode covering tectono-thermal events reflected by ages in the range 580–680 My. The pole from the Doornpoort Formation falls on the loop of McElhinny *et al.* close to poles of Middle Cambrian to Middle Ordovician age, consistent with a magnetization acquired in Damaran times.

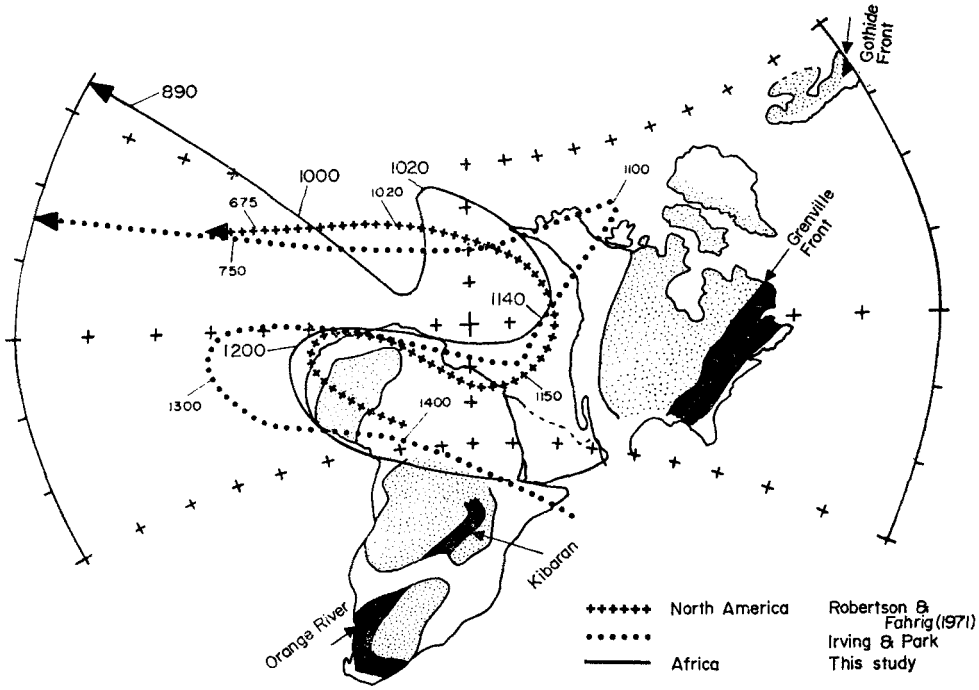


FIG. 16. Apparent polar wander curves for Africa and North America for the interval 1300–1000 My superimposed to show the distribution of Precambrian crust (stippled) and mobile belts belonging to this interval. The apparent polar wander curves shown in this figure are interpreted as south curves by McElhinny *et al* (1974).

4.3 *Intercontinental correlations between shield areas in Upper Proteroizoic times*

An attempt to reconstruct the positions of shield areas in Precambrian time was made by McElhinny *et al.* (1968) who found that their apw curve for the interval 1950–1310 My could be fitted to the then-available data from the Canadian shield for a similar interval. Subsequent data have altered both the shape and the age control of the apw curves from each shield area, and the reconstruction suggested by McElhinny *et al.* (1968) cannot now be substantiated. Brock & Piper (1972) using data from Africa available up to 1971 compared the shapes of the African and North American apw curves and demonstrated that it was not possible to superimpose the curves over their whole length. The implication of this conclusion is that relative movement between the regions must have taken place during at least part of Precambrian time.

With improved definition of the Precambrian apw curves for Africa and North America (Irving & Park 1972) it is possible to attempt a much more rigorous comparison. Two major points emerge: firstly both the African and North American apw curves show tight loops of nearly identical size between 1300 and 1000 My, and secondly the two curves have quite different form after about 1000 My so that there is no possibility of superimposing them in Upper Proterozoic time. If the two loops between 1300 and 1000 My are superimposed for the two regions areas of known Precambrian crust are brought into close proximity as shown in Fig. 16. The 1300–1100 My mobile belts (Grenville, Kibaran and Orange River) then lie on an arc close to a great circle. Palaeomagnetic data from the Fenno-Scandian shield are sparse but it appears that the Gothide orogenic line (*c.*1000 My) also belonged to this lineament, and Donaldson *et al.* (1973, Fig. 10) have made a palaeomagnetic reconstruction to demonstrate this point; this is included in Fig. 16 to demonstrate the alignment of the belts at the time of their formation.

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